

Spring Core and Spring Boot - Hands On

Hands on 1: Create a Spring Web Project using Maven

pom.xml (generated from start.spring.io)

```
<groupId>com.cognizant</groupId>
<artifactId>spring-learn</artifactId>
<version>0.0.1-SNAPSHOT</version>
<name>spring-learn</name>

<parent>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-parent</artifactId>
  <version>3.2.5</version>
</parent>

<dependencies>
  <dependency>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-web</artifactId>
  </dependency>
  <dependency>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-devtools</artifactId>
    <scope>runtime</scope>
    <optional>true</optional>
  </dependency>
</dependencies>
```

SpringLearnApplication.java

```
package com.cognizant.springlearn;

import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class SpringLearnApplication {

    private static final Logger LOGGER =
        LoggerFactory.getLogger(SpringLearnApplication.class);

    public static void main(String[] args) {
        LOGGER.info("Application starting");
        SpringApplication.run(SpringLearnApplication.class, args);
        LOGGER.info("Application started");
    }
}
```

Output - console on running SpringLearnApplication

```
.
/\ \ / _ _ ' _ _ ( _ ) _ _ \ _ _ \ _ _ \
( ( ) \ _ _ | _ _ | _ _ | _ _ \ _ _ \ _ _ \
\ \ / _ _ ) | _ _ | _ _ | _ _ \ _ _ \ _ _ \
' | _ _ | _ _ | _ _ | _ _ \ _ _ \ _ _ \
=====|_|=====|_|/=/=/=/=/
:: Spring Boot ::                (v3.2.5)
```

```
INFO 9532 --- [          main] c.c.springlearn.SpringLearnApplication : Application
starting
INFO 9532 --- [          main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on
port(s): 8080 (http)
INFO 9532 --- [          main] c.c.springlearn.SpringLearnApplication : Started
SpringLearnApplication in 2.847 seconds
INFO 9532 --- [          main] c.c.springlearn.SpringLearnApplication : Application started
```

Hands on 2: Spring Core - Load SimpleDateFormat from Spring Configuration XML

date-format.xml (src/main/resources)

```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
       xsi:schemaLocation="http://www.springframework.org/schema/beans
                           https://www.springframework.org/schema/beans/spring-beans.xsd">

    <bean id="dateFormat" class="java.text.SimpleDateFormat">
        <constructor-arg value="dd/MM/yyyy" />
    </bean>

</beans>
```

SpringLearnApplication.java - displayDate() method

```
public void displayDate() throws ParseException {
    ApplicationContext context = new ClassPathXmlApplicationContext("date-format.xml");
    SimpleDateFormat format = context.getBean("dateFormat", SimpleDateFormat.class);
    Date date = format.parse("31/12/2018");
    System.out.println(date);
}
```

Output

```
Mon Dec 31 00:00:00 IST 2018
```

Hands on 3: Spring Core - Incorporate Logging

application.properties

```
logging.level.org.springframework=info
logging.level.com.cognizant.springlearn=debug
logging.pattern.console=%d{yyMMdd} %d{HH:mm:ss.SSS} %-20.20thread| %5p| %-
25.25logger{25} | %25M | %m%n
```

SpringLearnApplication.java - displayDate() updated with logging

```
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;

private static final Logger LOGGER = LoggerFactory.getLogger(SpringLearnApplication.class);

public void displayDate() throws ParseException {
    LOGGER.info("START");
    ApplicationContext context = new ClassPathXmlApplicationContext("date-format.xml");
    SimpleDateFormat format = context.getBean("dateFormat", SimpleDateFormat.class);
    Date date = format.parse("31/12/2018");
    LOGGER.debug("{} ", date);
    LOGGER.info("END");
}
```

```
}
```

Output

```
240715|11:02:41.208|main          |INFO |SpringLearnApplication |
displayDate|START
240715|11:02:41.331|main          |DEBUG|SpringLearnApplication |
displayDate|Mon Dec 31 00:00:00 IST 2018
240715|11:02:41.332|main          |INFO |SpringLearnApplication |
displayDate|END
```

Hands on 4: Spring Core - Load Country from Spring Configuration XML

country.xml (src/main/resources)

```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://www.springframework.org/schema/beans
    https://www.springframework.org/schema/beans/spring-beans.xsd">

  <bean id="country" class="com.cognizant.springlearn.Country">
    <property name="code" value="IN" />
    <property name="name" value="India" />
  </bean>

</beans>
```

Country.java

```
package com.cognizant.springlearn;

import org.slf4j.Logger;
import org.slf4j.LoggerFactory;

public class Country {

    private static final Logger LOGGER = LoggerFactory.getLogger(Country.class);

    private String code;
    private String name;

    public Country() {
        LOGGER.debug("Inside Country Constructor");
    }

    public String getCode() {
        LOGGER.debug("Inside getCode");
        return code;
    }

    public void setCode(String code) {
        LOGGER.debug("Inside setCode");
        this.code = code;
    }

    public String getName() {
        LOGGER.debug("Inside getName");
        return name;
    }
}
```

```

    public void setName(String name) {
        LOGGER.debug("Inside setName");
        this.name = name;
    }

    @Override
    public String toString() {
        return "Country [code=" + code + ", name=" + name + "]";
    }
}

```

SpringLearnApplication.java - displayCountry() method

```

public void displayCountry() {
    LOGGER.info("START");
    ApplicationContext context = new ClassPathXmlApplicationContext("country.xml");
    Country country = context.getBean("country", Country.class);
    LOGGER.debug("Country : {}", country.toString());
    LOGGER.info("END");
}

public static void main(String[] args) {
    SpringApplication.run(SpringLearnApplication.class, args);
    SpringLearnApplication app = new SpringLearnApplication();
    app.displayCountry();
}

```

Output

```

240715|11:14:02.101|main          |INFO |SpringLearnApplication |
displayCountry|START
240715|11:14:02.147|main          |DEBUG|Country                  |
<init>|Inside Country Constructor
240715|11:14:02.151|main          |DEBUG|Country                  |
setCode|Inside setCode
240715|11:14:02.152|main          |DEBUG|Country                  |
setName|Inside setName
240715|11:14:02.166|main          |DEBUG|SpringLearnApplication   |
displayCountry|Country : Country [code=IN, name=India]
240715|11:14:02.167|main          |INFO |SpringLearnApplication   |
displayCountry|END

```

Hands on 5: Spring Core - Demonstration of Singleton Scope and Prototype Scope

SpringLearnApplication.java - displayCountry() modified to get bean twice

```

public void displayCountry() {
    LOGGER.info("START");
    ApplicationContext context = new ClassPathXmlApplicationContext("country.xml");
    Country country = context.getBean("country", Country.class);
    Country anotherCountry = context.getBean("country", Country.class);
    LOGGER.debug("Country : {}", country.toString());
    LOGGER.debug("country == anotherCountry : {}", (country == anotherCountry));
    LOGGER.info("END");
}

```

Output - default scope (singleton)

```

240715|11:21:09.402|main          |DEBUG|Country                  |
<init>|Inside Country Constructor

```

```

240715|11:21:09.406|main          |DEBUG|Country          |
setCode|Inside setCode
240715|11:21:09.407|main          |DEBUG|Country          |
setName|Inside setName
240715|11:21:09.409|main          |DEBUG|SpringLearnApplication |
displayCountry|Country : Country [code=IN, name=India]
240715|11:21:09.409|main          |DEBUG|SpringLearnApplication |
displayCountry|country == anotherCountry : true

```

country.xml - scope changed to prototype

```

<bean id="country" class="com.cognizant.springlearn.Country" scope="prototype">
  <property name="code" value="IN" />
  <property name="name" value="India" />
</bean>

```

Output - prototype scope

```

240715|11:24:51.220|main          |DEBUG|Country          |
<init>|Inside Country Constructor
240715|11:24:51.223|main          |DEBUG|Country          |
setCode|Inside setCode
240715|11:24:51.224|main          |DEBUG|Country          |
setName|Inside setName
240715|11:24:51.226|main          |DEBUG|Country          |
<init>|Inside Country Constructor
240715|11:24:51.226|main          |DEBUG|Country          |
setCode|Inside setCode
240715|11:24:51.227|main          |DEBUG|Country          |
setName|Inside setName
240715|11:24:51.229|main          |DEBUG|SpringLearnApplication |
displayCountry|Country : Country [code=IN, name=India]
240715|11:24:51.229|main          |DEBUG|SpringLearnApplication |
displayCountry|country == anotherCountry : false

```

Hands on 6: Spring Core - Load list of countries from Spring Configuration XML

country.xml - four countries and a list bean

```

<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://www.springframework.org/schema/beans
    https://www.springframework.org/schema/beans/spring-beans.xsd">

  <bean id="in" class="com.cognizant.springlearn.Country">
    <property name="code" value="IN" />
    <property name="name" value="India" />
  </bean>

  <bean id="us" class="com.cognizant.springlearn.Country">
    <property name="code" value="US" />
    <property name="name" value="United States" />
  </bean>

  <bean id="de" class="com.cognizant.springlearn.Country">
    <property name="code" value="DE" />
    <property name="name" value="Germany" />
  </bean>

  <bean id="jp" class="com.cognizant.springlearn.Country">

```

```

        <property name="code" value="JP" />
        <property name="name" value="Japan" />
    </bean>

    <bean id="countryList" class="java.util.ArrayList">
        <constructor-arg>
            <list>
                <ref bean="in"></ref>
                <ref bean="us"></ref>
                <ref bean="de"></ref>
                <ref bean="jp"></ref>
            </list>
        </constructor-arg>
    </bean>

</beans>

```

SpringLearnApplication.java - displayCountries() method

```

@SuppressWarnings("unchecked")
public void displayCountries() {
    LOGGER.info("START");
    ApplicationContext context = new ClassPathXmlApplicationContext("country.xml");
    List<Country> countryList = (List<Country>) context.getBean("countryList");
    LOGGER.debug("Countries : {}", countryList);
    LOGGER.info("END");
}

```

Output

```

240715|11:31:47.018|main          |INFO |SpringLearnApplication |
displayCountries|START
240715|11:31:47.052|main          |DEBUG|Country                |
<init>|Inside Country Constructor
240715|11:31:47.055|main          |DEBUG|Country                |
setCode|Inside setCode
240715|11:31:47.055|main          |DEBUG|Country                |
setName|Inside setName
240715|11:31:47.057|main          |DEBUG|Country                |
<init>|Inside Country Constructor
240715|11:31:47.057|main          |DEBUG|Country                |
setCode|Inside setCode
240715|11:31:47.058|main          |DEBUG|Country                |
setName|Inside setName
240715|11:31:47.059|main          |DEBUG|Country                |
<init>|Inside Country Constructor
240715|11:31:47.059|main          |DEBUG|Country                |
setCode|Inside setCode
240715|11:31:47.060|main          |DEBUG|Country                |
setName|Inside setName
240715|11:31:47.061|main          |DEBUG|Country                |
<init>|Inside Country Constructor
240715|11:31:47.061|main          |DEBUG|Country                |
setCode|Inside setCode
240715|11:31:47.062|main          |DEBUG|Country                |
setName|Inside setName
240715|11:31:47.065|main          |DEBUG|SpringLearnApplication |
displayCountries|Countries : [Country [code=IN, name=India], Country [code=US, name=United
States], Country [code=DE, name=Germany], Country [code=JP, name=Japan]]
240715|11:31:47.065|main          |INFO |SpringLearnApplication |
displayCountries|END

```